MIL-P-2496D (MU)
9 January 1967
SUPERSEDING
MIL-P-2496C (MU)
12 February 1964
(se 6.11)

MILITARY SPECIFICATION

PRIMER, PERCUSSION, M29A1

PARTS AND LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the parts and the loading, assembling, and packing for one type of initiating device designated as Primer, Percussion, M29A1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitations for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 — Ammunition and Special Weapons; General Specification for.

MIL-I-45607 — Inspection Equipment, Supply and Maintenance of.

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105).

MIL-STD-109 — Quality Assurance Terms and Definitions.

MIL-STD-650 — Explosives, Sampling, Inspection, and Testing.

MIL-STD-1168 — Lot Numbering of Ammunition.

MIL-STD-1235 — Single and Multi-level Continuous Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

U.S. ARMY MUNITIONS COMMAND

8798312 — Primer, Percussion, M29A1, Assembly

8857391 — Box, Packing, Ammunition.

FSC 1375

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For, Primer, Percussion, M29A1 or #26.

8857392 — Box, Set-up, Packing, Ammunition, Inner and Outer for Primer, Percussion, M29A1 or #26.

PUBLICATIONS

ET 8798312 — Equipment Tabulation Number.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Assembly. The percussion primer assembly shall comply with all requirements specified in Drawing (dwg.) 8798312 and with all requirements specified in applicable specifications.

3.3 Moisture content of composition. At the time of loading, the moisture content of the composition used in loading the primer shall not exceed 0.30 percent max. when tested as specified in 4.4.1.

3.4 Functioning.

3.4.1 *Sensitivity.* The primer shall fire on the first attempt without squibbing (see 6.8) or hangfire under an applied energy of 15.68 inch ounces (see 6.7) maximum (max.) but shall not fire under an applied energy of 1.92 inch ounces minimum (min.) when tested as specified in 4.4.2.1.

3.4.2 *Failure to fire (Dud).* The primers shall fire on the first attempt without indication of audible hangfire when tested as specified in 4.4.2.2.

3.5 Workmanship. The parts and assemblies shall be free of chips, dirt, grease, stains, corrosions, loose explosive and other foreign material. The cleaning method shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined.

3.6 First article inspection. This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions. Unless otherwise specified in the contract, purchase order, or this specification the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL–STD–109 in order to define the terms used herein. The provisions of MIL–A–2550 shall apply.

4.1.1 *Submission of product.* At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

(a) A statement that the lot complies with all quality assurance provisions specified within this specification.

(b) Number of units of product inspected.

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(c) Results obtained, by defect code, for all inspections performed.

(d) Drawing, specification number and date, together with an identification and date of changes.

(e) Statements of findings on the composition and certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.

(f) Number of items in the lot.

(g) Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.2 *Government verification.* Using the contractor's written quality assurance, procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 **First article inspection.**

4.2.1 *Submission.* The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with the provisions of 4.2.2. The first article sample shall consist of 50 complete inert sets of parts (i.e. 50 each of every inert part) which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials, (see 4.1.1(e)). A sample containing known defects will not be submitted unless authorized by the contracting officer. A first article sample, or portion thereof, as directed by the contracting officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing, or specification such as to significantly affect product uniformity as determined by the Government.

4.2.2 *Inspection to be performed.* The parts will be subjected by the Government to any or all of the applicable examinations specified in 4.3 of this specification and any or all requirements of the applicable drawings.

4.2.3 *Rejection.* If any part fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of a part to comply with any of the stated requirements.

4.3 **Inspection provisions.**

4.3.1 *Lot formation.* The term "lot" as

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used throughout this specification refers to an inspection lot, which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, specification, or revision thereof. Changes to either the process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government inspector, shall not be deemed to alter the homogeneity of the lot. Each lot shall contain:

(a) Parts of one kind from one manufacturer.

(b) Composition produced by one manufacturer under one continuous set of operating conditions and which consists of one or more batches that have been subjected to the same unit chemical or physical mixing process intended to make the final product homogenous.

Inspection lots shall comply with MIL–STD–105 and shall be numbered in accordance with MIL–STD–1168.

Unless otherwise approved by the contracting officer, the inspection lot size of major assemblies or end items deliverable under the contract shall be not less than the smallest weekly estimate of quantities contractually scheduled for production during the contract period nor more than the largest quantity contractually scheduled for delivery during any month of the contract period. Inspection lots for components or sub-assemblies, other than the items of delivery, shall be homogeneous and of a size convenient to the contractor.

4.3.2 *Examination.* Sampling plans and procedures for the following classification of defects shall be in accordance with MIL–STD–105 except that inspection for critical defects shall be 100 percent. Contractor's sampling plans, if used, shall be approved by the Government and shall provide, as a minimum, the protection afforded the Government by the sampling plans in MIL–STD–105. Continuous sampling plans in accordance with MIL–STD–1235 for defects other than critical may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 *Cup* (see dwg. 8798313 covering a detail of dwg. 8798312).

Categories	*Defects*	*Method of Inspection*	*Code No. (see 6.2)*
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Thickness through bottom	Gage	01001
Minor:	AQL 1.50 percent		
201.	Outside diameter, minimum (min.)	Gage	01002
202.	Inside diameter, maximum (max).	Gage	01003
203.	Length, min.	Gage	01004
204.	Evidence of poor workmanship (see 3.5)	Visual	01005

4.3.2.2 *Anvil* (see dwg. 8798314 covering a detail of dwg. 8798312).

Categories	*Defects*	*Method of Inspection*	*Code No.*
Critical:	None defined		

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Categories	Defects	Method of Inspection	Code No.
Major:	AQL 1.00 percent		
101.	Thickness of flange	Gage	02001
102.	Total height	Gage	02002
103.	Diameter	Gage	02003
104.	Diameter of flash holes	Gage	02004
105.	Radius incorrect	Gage	02005
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.5)	Visual	02006

4.3.2.3 *Assembly, primer, prior to inserting cover* (see dwg. 8798312).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Charge weight incorrect	Balance	03001
102.	Charge missing	Visual	03002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.5)	Visual	03003

4.3.2.4 *Assembly, primer, prior to assembling anvil* (see dwg. 8798312).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Cover missing or improperly positioned	Visual	04001
102.	Double or multiple cover	Visual	04002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.5)	Visual	04003

4.3.2.5 *Assembly, primer* (see dwg. 8798312).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Cup split, cracked, or punctured	Visual	05001
Major:	AQL 1.00 percent		
101.	Length, max.	Gage	05002
102.	Diameter	Gage	05003
103.	Anvil loose, cocked, or inverted	Visual-Manual	05004
104.	Vent hole missing in anvil	Visual	05005
105.	Primer damaged to extent that function may be impaired	Visual	05006
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.5)	Visual	05007

4.3.2.6 *Box, set-up, inner and outer, prior to sealing* (see dwg. 8857392).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Incorrect number of primers	Visual	06001
Minor:	AQL 1.50 percent		
201.	Tray damaged	Visual	06002
202.	Tray not glued to bottom of box	Visual-Manual	06003
203.	Evidence of poor workmanship (see 3.5)	Visual	06004
204.	Inner cartons are loose in outer cartons	Manual	06005

4.3.2.7 *Box, set-up, inner and outer, sealed* (see dwg. 8857392).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		

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Categories	Defects	Method of Inspection	Code No.
	101. Sealing tape torn, incomplete or insecure	Visual-Manual	07001
Minor:	AQL 1.50 percent		
	201. Box damaged to extent that contents are exposed or liable to become exposed	Visual	07002
	202. Marking misleading or unidentifiable	Visual	07003
	203. Label missing, incompletely sealed, or improperly positioned	Visual	07004
	204. Evidence of poor workmanship (see 3.5)	Visual	07005

4.3.2.8 *Sealed barrier bag* (see dwg. 8857392).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
	101. Seal incomplete or incorrectly positioned	Visual	08001
	102. Bag punctured, torn, or cut	Visual	08002
Minor:	None defined		

4.3.2.9 *Packing box, prior to sealing* (see dwg. 8857391).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
	101. Set-up box missing	Visual	09001
Minor:	AQL 0.65 percent		
	201. Packing component missing	Visual	09002

4.3.2.10 *Sealed packing box* (see dwg. 8857391).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
	101. Box damaged to extent that contents are exposed or liable to become exposed	Visual	10001
	102. Strapping or hardware missing, broken, or loose	Visual-Manual	10002
	103. DOD symbol missing or incorrect	Visual	10003
Minor:	AQL 1.50 percent		
	201. Car seal missing, unsealed, or incorrectly positioned	Visual-Manual	10004
	202. Marking misleading or unidentifiable	Visual	10005
	203. Contents loose	Visual	10006
	204. Strapping or hardware improperly engaged or assembled	Visual-Manual	10007

4.3.3 *Testing.*

4.3.3.1 *Moisture content of composition* (see 3.3). Major defect—Code No. 11001. A sample of approximately one gram shall be taken from each container of composition as it arrives at the loading station and at 4-hour intervals thereafter with a final sample taken at the end of the loading day. The percentage of moisture shall be determined as specified in 4.4.1. When a sample is found to be nonconforming, loading from the container shall be stopped and all primers loaded since the last acceptable determination shall be rejected. The remaining contents of the container may be subjected to corrective action but a verification sample shall be selected, tested, and found to be acceptable prior to loading again from that container.

4.3.3.2 *Functioning* (see 3.4).

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4.3.3.2.1 *Sensitivity* Major defect—Code No. 05008. A sample of 400 percussion primers shall be selected from each lot for this test. The test shall be performed as specified in 4.4.2.1, using test equipment specified in 4.3.4. If the sum of the average critical height (H average) plus 5 times the standard deviation (S) multiplied by the weight of the ball exceeds the maximum requirement or if the sum of the average critical height (H average) minus 2 times the standard deviation (S) multiplied by the weight of the ball is less than the minimum requirement, the lot shall be rejected. The standard deviation shall be calculated using the formula contained in 6.6.

4.3.3.2.2 *Duds*—Major defect—Code No. 05009. A sample of 300 percussion primers shall be selected from each lot for this test. The failure of two or more primers to function during this test shall be cause for rejection of the entire lot. The test shall be performed as specified in 4.4.2.2 using test equipment specified in 4.3.4.

4.3.3.3 *Determination of composition* (see dwg. 8798312)—Major defect—Code No. 11002. A sample of approximately one gram shall be selected from each batch or blend of composition and tested as specified in 4.4.3. The batch or blend of batches shall be rejected if one or more of the requirements are not met.

4.3.4 *Inspection Equipment.* Equipment Tabulation Number ET–8798312 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3.

4.3.4.1 *Government rights to documentation.* Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey, and whenever inspection equipment drawings are needed.

4.3.4.2 *Supply and maintenance.* Supply and maintenance of the equipment listed in the ET shall be in accordance with MIL–I–45607.

4.3.4.3 *Government use of contractors inspection equipment.* The contractor shall make available to the Government all inspection equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.4 Test methods and procedures.

4.4.1 *Moisture content of composition.* The moisture content of the composition shall be determined in accordance with Method 101.6 of MIL–STD–650. The sample used for this moisture determination shall be retained for use in the composition determination.

4.4.2 *Functioning.* Prior to performing these tests the primers shall be conditioned at a temperature of 75 degrees Fahrenheit (F) plus or minus 5 degrees F., for a period of 24 hours min.

4.4.2.1 *Sensitivity.* The primers shall be divided into 8 groups of 50 primers each and tested in the following manner: The first group (50) of primers shall be tested by dropping a steel ball weighing 1.94 plus or minus .02 ounces on the firing pin of the test fixture from a height of 5 inches. (The height of drop shall be measured as the height from the top of the firing pin to the bottom of the ball.) The number of primers that fire, fail to fire, hangfire, or squib shall be recorded. This procedure shall be repeated as successively increasing heights of 1 inch, using a new group of 50 primers at each different

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height, until a height is reached at which all primers in the group being tested fail to fire. Acceptance or rejection of the lot shall be determined from calculations based on recorded data (see 6.6). All samples subjected to test shall be scrapped. Any remaining untested samples may be returned to the lot from which they were taken.

4.4.2.2 *Duds.* Each of the 300 sample primers shall be tested by dropping a steel ball weighing 1.94 plus or minus .02 ounces on the firing pin of the test fixture from a height not to exceed 8 inches.

4.4.3 *Determination of composition.* The dried sample from the moisture determination of composition shall be subjected to the following composition analysis procedure:

4.4.3.1 *Barium nitrate.* An accurately weighed portion of approximately one gram of the composition shall be transferred to a tared, 30 milliliter (ml.) fritted glass crucible of medium porosity. Three ml. of distilled water, at a temperature of 5 plus or minus 2 degrees C. shall be added. This mixture shall be agitated for one minute with a rubber policeman attached to a glass rod, breaking up any lumps of material which may be present. Suction shall be applied and the alternate extraction and suction procedure shall be repeated 5 times. The residue of ethyl alcohol, aspirated until the alcohol odor is not detectable then dried in an oven, maintained at 60 plus or minus 5 degrees C. for 30 minutes. The crucible and contents shall be cooled in a desiccator and weighed. The crucible and residue shall be retained for the determination of basic lead styphnate. The percentage of barium nitrate on the moisture free basis shall be calculated in accordance with the following formula:

Percent of barium nitrate = (W-A) 100 divided by W where:

W = weight of composition taken
A = weight of residue after aqueous extraction

4.4.3.2 *Basic lead styphnate.* To the residue obtained in the determination of barium nitrate, 5 ml. of a saturated solution of ammonium acetate at approximately 25 degrees C. shall be added. Hot ammonium acetate solution shall not be used since it dissolves antimony sulfide to some extent. The mixture in the crucible shall be agitated for approximately 30 seconds with a rubber policeman to break up any agglomerates, and the extracting solution shall be permitted to remain in contact with the residue for an additional 90 seconds. The extraction shall be repeated 5 times with saturated ammonium acetate solution. The residue shall be washed with alternate portions of distilled water, cooled to 5 plus or minus 2 degrees C., and saturated ammonium acetate solution at approximately 25 degrees C., until both the residue and the filtrate are colorless. The residue shall then be washed throughly with cold distilled water and 3 portions of ethyl alcohol. The crucible and contents shall then be dried in an oven maintained at 60 plus or minus 5 degrees C. for 30 minutes, then cooled in a desiccator and weighed. The loss in weight of the residue shall be calculated to the percentage of basic lead styphnate on a moisture free basis in accordance with the following formula:

Percent of basic lead styphnate = 100 (B–C) divided by W where:

B = weight of residue and crucible after extraction of barium nitrate.

C = weight of residue and crucible after extraction of basic lead styphnate.

W = weight of composition taken.

4.4.3.3 *Tetracene.* The residue obtained from the determination of basic lead styphnate shall be quantitatively transferred to a 150 ml. beaker with a stream of distilled water from a wash bottle. Fifty ml. of distilled water shall be added to the beaker, and the beaker and contents shall be boiled on a hot plate for 5 minutes. The contents of the beaker shall be filtered through the original

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crucible. Any residue remaining in the beaker shall be quantitatively transferred to the filtering crucible with 3 portions of boiling distilled water. The residue in the crucible shall be washed with 3 small portions of ethyl alcohol. The crucible shall be dried in an oven maintained at 60 plus or minus 5 degrees C. for 30 minutes. The crucible and contents shall be cooled in a desiccator and weighed. The loss in weight shall be calculated as the percentage of tetracene on a moisture free basis, in accordance with the following formula:

Percent tetracene = 100 (C-D) divided by W where:

C = weight of residue and crucible after extraction of basic lead styphnate.
D = weight of residue and crucible after extraction of tetracene.
W = weight of composition taken.

4.4.3.4 *Antimony sulfide.* The weight of residue obtained after the determination of tetracene shall be calculated as percentage of antimony sulfide on a moisture free basis, in accordance with the following formula:

Percent of antimony sulfide = 100 (D — E) divided by W where:

D = weight of residue plus crucible after extraction of tetracene.

E = weight of crucible.

W = weight of composition taken.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 *Level C.* The primers shall be preserved and packaged in accordance with dwg. 8857392.

5.2 Packing.

5.2.1 *Level C.* The sealed boxes shall be packed in accordance with dwg. 8857391.

6. NOTES

6.1 **Ordering data.** Procurement documents shall specify the following:

(a) Title, number and date of this specification.

(b) Data cards. Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

(c) Submission of first article quantity. Instructions as to the location of the facility at which the evaluation of the first article quantity shall be carried out, may be obtained from the contracting officer.

(d) Primers procurred as commercial substitutes for the M29A1 Primer, as specified in Note 14 of dwg. 8798312, shall be required to meet only the functioning characteristics as defined by 3.4 and examinations as specified by 4.3.2.5.

6.2 Inspection code numbers. The five-digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment. The contractor shall design inspection equipment as required by the referenced Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.7.

6.3.1 *Equipment lists.* Equipment part or identifying lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (SMUPA Form 1010) the numbers of drawings of existing equipment designs or codes as indicated in paragraph 6.3.2. Design action required of the contractor with respect to the different types of drawings that may be listed is described in paragraphs 6.3.3 and 6.3.4. Action required

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by the contractor with respect to commercial inspection equipment is described in 6.3.5. The contractor will be required to prepare detailed drawings in accordance with 6.3.6 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.2 *Inspection equipment list codes.* The inspection equipment as defined in 6.3.3, 6.3.5, and 6.3.6 will be designated in the EL by the following codes. Other inspection equipment listed on the EL and not coded are defined in 6.3.4.

MU — Army design, mandatory for use.

CD — Contractor design and/or commercial equipment on contractor format. (Requires submission for Government approval as specified in 6.3.7).

6.3.3 *Army designs.* Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications, mandatory and non-mandatory. When an equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs (see 6.3.6) or commercial equipment selections (see 6.3.5) must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.4 *Specification control drawing.* Specification control drawings depict the minimum equipment requirements in outline, descriptive, diagrammatic or pictorial form only and specify the required performance or other characteristics. Contractors must prepare detailed drawings (see 6.3.6) of their designs in support of specification control drawings. These contractor prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Commercial equipment meeting the requirements of specification control drawings must be described in sufficient detail to permit identification and evaluation by the Government (see 6.3.5). Designs shall be submitted for approval as specified in 6.3.7.

6.3.5 *Commercial equipment.* Commercial equipment is inspection equipment that has universal application for a specific function. It is comprised of items commonly used by industry and Government. Contractors normally are not required to furnish drawings of commercial inspection equipment, but a list of such equipment must be approved by the Government. When commercial inspection equipment requires the use of special fixturing, the design of this fixturing shall be considered as part of the total design, and 6.3.6 shall apply. In the latter case, only those portions of the commercial equipment need be detailed in the designs as are necessary to completely fabricate, calibrate, and operate the total item of inspection equipment. Lists and fixturing designs shall be submitted for approval as specified in 6.3.7.

6.3.6 *Contractor designs.* Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate, and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly de-

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picted in accordance with approved practices to the extent that further calculation or clarification will not be required. Contractor designs identified as CD may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above. Contractor designs identified as CD shall comply with the format and requirements of MIL-D-45608, and in addition, contain the detail and information specified above. Final approved contractor designs identified as CD shall be suitable for microfilming. They must be non-reproducable diazo or electrostatic prints of top quality full size with black lines on white background and shall provide a legible print produced via the 35mm microfilm process. Contractors shall submit one copy of the final design as a rolled set or flat set. Designs shall be submitted for approval as specified in 6.3.7.

6.3.7 *Submission of contractor designs.* Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the Contracting Officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

6.4 Submission of test data and ammunition data cards. In addition to the normal distribution of records, for loaded items procured by the Department of the Army, one copy of all functioning test data and ammunition data cards shall be forwarded to the following:

Commanding Officer
Picatinny Arsenal
Dover, New Jersey
ATTN: SMUPA-ND5
SMUPA-DE

6.5 Check test for possible deterioration. The following test is designed to prevent the loading of primers which have deteriorated into a major ammunition item. The test should be performed on primers from each lot which have been stored under normal conditions for more than 2 years or under adverse conditions for any period of time. If the specification for the ammunition item into which the primers are loaded does not require that a check test be performed, the contracting officer should insert in the contract a provision for the performance of the check test.

6.5.1 *Sensitivity.* Code No. 05008. The test shall be performed as specified in 4.4.2.1, using test equipment in accordance with 4.3.4. A sample of 600 primers shall be selected from each lot for test. Seventy-five primers shall be tested at each height. If H average plus 4.66 S exceeds 8 inches, or if H average minus 1.86 S is less than 1 inch, the lot shall be rejected.

6.5.2 *Duds.* This test shall be performed as specified in 4.4.2.2. A sample of 300 primers shall be selected from each lot for test. If three or more primers fail to function on the first attempt, the lot shall be rejected.

6.6 Formulae and method for determining H average and S.

6.6.1 *Formulae:*

H average = A plus B plus C

S = I times the square root of L minus M

where:

A = the sum of Column 3.

B = the first drop height at which all primers misfired.

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C = one half the drop height increment (see 6.9).

L = the sum of Column 5.

M = the square of the sum of Column 3.

I = the drop height increment.

6.6.2 *Tabulation of data.* Divide a sheet of paper into 5 columns and tabulate the data as follows:

(a) Title Column 1 "Height of Drop" and list all drop heights, in consecutive order starting with the lowest height. Do not list the heights at which all elements fired or misfired.

(b) Title Column 2 "Number Misfired" and list the sum of the numbers of elements that misfired and "squibed" at each intermediate height.

(c) Title Column 3 "Decimal Fraction of Elements Failing to Fire at Each Intermediate Height" and record the decimal fraction that misfired at each intermediate height. This number is obtained by dividing the value obtained under "Number Misfired" by the number of elements tested at that height. Calculation should be carried out and recorded to the nearest one hundredth (.01).

(d) Title Column 4 "Standard Deviation Factor" and list odd numerals, in sequence starting with 1, for as many drop heights as are listed in Column 1.

(e) Title Column 5 "Standard Deviation Factor times Fraction Misfired" and, by multiplying each decimal fraction in Column 3 by the number on the same line in Column 4, enter the product on the corresponding line in Column 5.

6.6.3 *Procedure for solving equations.*

(a) To obtain "A"—add the values in Column 3 and substitute the sum for "A".

(b) To obtain "B"—substitute the first drop height at which all elements misfired for "B".

(c) To obtain "C"—divide the drop height increment by 2 and substitute the dividend for "C".

(d) To obtain "L"—add the values in Column 5 and substitute the sum for "L".

(e) To obtain "M"—square the sum of Column 3 and substitute the product for "M".

(f) To obtain "H average"—add the values substituted for A, B, and C.

(g) To obtain "S"—substract the value substituted for "M" from the value substituted for "L" and then obtain the square root of the difference, and then multiply the square root by the height increment.

(h) Using the calculated values for "H average" and "S", if "H average" plus 5 "S" multiplied by the max. weight of the ball is equal or less than the max. inch ounce energy value required, and "H average" minus 2 "S" multiplied by the min. weight of the ball is equal or greater than the min. energy value required, the lot should be considered acceptable for sensitivity.

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6.7 Inch ounces. The product of drop height in inches multiplied by the weight of the ball in ounces.

6.8 Squib. Defined as a burning of the composition without detonation.

6.9 Height increment. The height increment is defined as a constant interval in inches between successive drop heights which must be considered in calculating average height and the standard deviation in drop height, whenever it differs from one inch.

6.10 Check test expense. The contracting officer should arrange to have the contractor reimbursed for any expense incurred in the performance of the check test for possible deterioration.

6.11 Supersession data. This specification includes the requirements of MIL-P-46544A (MU) dated 16 December 1963.

Custodian:

Army—MU

Preparing activity:

Army—MU

Project No. 1375-A-1051

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT

$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 OCT 64 1426

ESC-FM 2011-66

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